

Bridging Artistic Expression and Game Design through Cross-Cultural Collaboration: An Educational Framework from a Seven-Year Japan–U.S. Program

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Figure 1: Examples of Japan–U.S. Collaboration works by students.

Abstract

Contemporary game development, particularly within the expanding independent game scene, increasingly requires small teams closely integrate artistic expression with system-oriented game design. However, in many educational contexts, these competencies are cultivated through different pedagogical emphases, creating tension when students attempt to integrate creative experiment with structured design processes focused on formal systems of rules, mechanics, and player interaction. This paper examines how cross-cultural education can function as a pedagogical framework for bridging such differing perspectives. Focusing on a seven-year international collaboration between an art university in Japan and a game program at a U.S. university, we report on an educational program designed to support collaboration between expression-oriented students and game design students who emphasize system design, including mechanics. Rather than assigning fixed disciplinary roles, the program foregrounded cross-disciplinary “translation,” encouraging students to articulate artistic intentions, negotiate technical constraints, and reflect on differing assumptions about what constitutes a “good” or “finished” game. Based on qualitative observations from workshops, collaborative game development, and public exhibitions, this study identifies three educational strategies that supported the bridging process: (1) team structures that prioritize disciplinary translation over role division, (2) early-stage

design frameworks that foreground expressive intent and the design of player interaction, and (3) the intentional use of cross-cultural differences as educational resources. These findings suggest that international collaboration can function not merely as a site of exchange, but as an effective educational framework for integrating artistic expression and system-oriented game design, particularly within art-based game education.

CCS Concepts

• **Applied computing** → **Education**; *Computer games*; • **Human-centered computing** → *Collaborative learning*.

Keywords

Game, Art, Education, International collaboration

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1 Introduction

The development of the independent game scene has changed the aesthetic paradigms of game design and created a demand for games that bring together artistic expression and experimental systems design [Juul 2020]. Expressive intent, mechanics, rules, and player experience are no longer addressed through clearly separated roles, but are negotiated and adjusted in parallel throughout the development process. Games have been described as media in which formal



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structures such as rules and mechanics are inseparably linked with the generation of meaning and experience, making expression and systems fundamentally interdependent [Tekinbaş and Zimmerman 2004]. In educational contexts, however, artistic exploration and system-oriented design are often cultivated under different pedagogical emphases. Game design is often taught as part of technical disciplines, such as computer science, with a focus on technical and highly structured design procedures. Meanwhile, artistic exploration is mostly taught in fine arts programs with a focus on experiment and free-form creative development. As the philosophies informing free-form experimentation and structured design procedures are quite different and often do not coexist, students tend to encounter difficulties trying to work across the two areas of activity to adopt an integrated and harmonic work process that satisfies both aspects of game creation [Card et al. 2021]. In art university environments, visual expression and concept formation are frequently foregrounded, while opportunities to engage with games as formal systems vary depending on institutional structures and curricular focus. Cross-cultural collaboration does not generate this tension anew; rather, it makes existing differences in values, communication styles, and educational assumptions visible. When students from different backgrounds collaborate, they often bring different priorities, which makes implicit criteria—such as what counts as a good game, or when a work can be considered finished—explicit and open to discussion within the team. Drawing on an seven-year collaboration between an art university in Japan and a game program at a U.S. university, this paper examines how cross-cultural education can function as a framework for bridging artistic expression and system-oriented game design.

2 Educational Context and Program Design

Art schools and game-focused programs often operate within distinct educational cultures. While art universities tend to emphasize artistic exploration, conceptual development, and visual expression, game education programs commonly foreground mechanics, rules, and player interaction. When students from these different backgrounds collaborate, disciplinary and cultural assumptions intersect, generating productive tension around how expressive intent can be realized as a playable system. The collaboration examined in this study connected Tokyo University of the Arts and the USC Games program, bringing together participants situated within differing cultural and educational contexts in Japan and the United States. Students from Tokyo University of the Arts typically came from majors such as animation, film, and media art, and had received education that prioritized visual expression, narrative construction, and concept development. Although many had extensive experience playing games, their exposure to game development as a design practice structured around rules and mechanics was often limited. In contrast, students from the U.S. program commonly had experience with core game development practices such as prototyping, mechanics design, and playtesting, while their levels of engagement with artistic expression varied across individuals. Rather than functioning as a one-time exchange, the program was designed as a recurring educational framework that combined workshops, student exchanges, collaborative game development, and public exhibitions within an annual cycle. Fixed role assignments

such as “artist” or “programmer” were intentionally avoided. Instead, students negotiated responsibilities dynamically in response to emerging ideas and technical constraints, repeatedly reconsidering who should take on which tasks as projects evolved. This design prioritized cross-disciplinary translation over division of labor, with the aim of making differing assumptions visible within the collaborative process. The process of cross-disciplinary translation became a core characteristic of the pedagogical approach of this course. Through the process of verbalizing and translating between different ways of perceiving, thinking, and expressing, students were able to develop an understanding of the aspects of the process of game-making and extend their creative means. In its earlier phases, the program employed long-term collaboration spanning several months, while later years introduced shorter, intensive formats in the form of game jams. As students and faculty gradually developed shared vocabularies for discussing aesthetics, mechanics, and play experience, essential bridging processes can be activated within increasingly compressed timeframes. Across both formats, public exhibitions played a consistent role, prompting reflection on how expressive intent and system design were communicated to audiences beyond the development team.

3 Bridging Strategies

The educational objective of this program was positioned as fostering awareness, understanding, and openness through an active process of translation between artistic expression and system-oriented perspectives. Through this process, students were encouraged to incorporate what they learned and integrate it into a renewed and unified practice toward expressive and self-directed independent game creation. A central feature of this approach was role fluidity. As projects progressed, students moved between expressive, systemic, and organizational responsibilities, depending on emerging ideas and constraints. Through this movement, differing assumptions—such as who is considered an author, how decisions should be made, and what constitutes progress—became visible within the production process rather than remaining implicit. Within this framework, three strategies supported the bridging process. First, team structures were designed to prioritize cross-disciplinary translation over role division, requiring students to explain and reinterpret ideas across expressive and systemic perspectives. Second, in collaborative game development, design frameworks foregrounded expressive intent and the design of player interaction rather than technical completeness, positioning these elements as shared goals for the entire production. Third, cross-cultural differences were intentionally treated as educational resources, making variations in values, communication styles, and evaluation criteria explicit and available for reflection. Together, these strategies conceived bridging as an active and ongoing learning process that negotiates between different ways of perceiving, thinking and expressing, rather than a way of realizing a final product.

4 Implementation and Observations

This study draws on multiple collaborative workshops and game development projects conducted over a seven-year period. In each iteration, small mixed teams of students from Japan and the United States engaged in exploratory game production (Figure. 1) and

presented their work publicly. Observations focused not on technical completeness or visual polish, but on how students exchanged ideas and articulated intentions during the collaborative process. During workshops, a design-oriented perspective on games was introduced, accompanied by shared vocabulary for understanding games as structures of possibilities for action and player experience. This vocabulary subsequently served as a common reference point in collaborative production. Over time, noticeable shifts emerged in how students discussed their work. Students with a stronger orientation toward artistic expression increasingly referred to systems and player interaction, while those with a stronger system-oriented background became more attentive to expressive intent, atmosphere, and ambiguity. Rather than converging toward a single mode of thinking, students developed an increased capacity to translate between perspectives when making design decisions. Moments of friction—such as disagreements over project scope, goals, or levels of completion—made educational and cultural assumptions particularly visible. In these situations, cross-cultural collaboration prompted students to articulate implicit criteria and reconsider priorities that might otherwise have remained unexamined. The transition to shorter, intensive formats further clarified the role of shared vocabulary in the bridging process, demonstrating that once communicative practices were established, similar forms of negotiation could occur within compressed timeframes. Public exhibitions visualized gaps between intention and reception, encouraging reflection and reconsideration of design decisions. Figures 2 and 3 show examples of collaborative team activity and public exhibitions, respectively.



Figure 2: Examples of collaborative team activity.

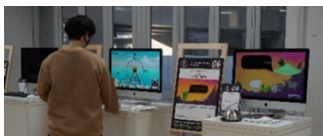


Figure 3: Examples of public exhibitions.

5 Discussion

The findings of this study suggest that cross-cultural collaboration functioned as a learning condition that made disciplinary assumptions visible and open to discussion. Rather than resolving tensions between artistic expression and system-oriented design, the program treated such tensions as opportunities for learning. By encouraging students to articulate and negotiate differing perspectives, collaboration became a site where expressive ambition and systemic realization could be examined in relation to one another. In contrast to international education initiatives that primarily

emphasize institutional frameworks or demonstrable outcomes, this study focused on micro-level interactions within small teams. Instead of evaluating collaboration itself as a success metric, the analysis examined the pedagogical mechanisms operating inside collaborative processes—specifically how translation, negotiation, and re-articulation across disciplinary and cultural boundaries supported learning. The role of time also emerged as a significant consideration. While long-term collaboration allowed for gradual alignment and reflection, the introduction of shorter, intensive formats demonstrated that bridging did not depend solely on duration. Rather, once shared vocabularies and communicative practices were established, similar forms of negotiation could occur within compressed timeframes. From this perspective, the educational value of long-term collaboration lies less in sustained production than in the accumulation of shared practices for discussing aesthetics, mechanics, and play experience. This study is situated within a specific institutional and cultural context and does not propose a universal model for game education. Nevertheless, it offers a set of transferable design principles that may inform other interdisciplinary and international educational settings. By framing difference—not as a problem to be minimized, but as a resource for learning—this approach highlights the pedagogical potential of cross-cultural collaboration in art-based game education.

6 Conclusion

This paper examined how cross-cultural and interdisciplinary collaboration can bridge artistic expression and system-oriented game design, drawing on seven years of collaborative practice between Japan and the United States. By framing collaboration as a process of translation across disciplinary and cultural boundaries, the program created learning environments in which students could move back and forth between expressive ambition and systemic realization while making design decisions. Through strategies such as role fluidity, design frameworks that position expressive intent and interaction design as shared goals, and the pedagogical use of cross-cultural differences, bridging emerged not as a fixed outcome but as an ongoing learning practice. While this study does not propose a universal model, it demonstrates the value of treating difference as an educational resource and offers transferable principles that may inform interdisciplinary and international approaches to game education in other institutional contexts.

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